

Computer Architecture and Organization
Written Assignment

1- Discuss asynchronous data transfer methods.

Asynchronous data transfer methods usually signal the start and end of the data being transmitted as opposed to Synchronous ones. In a sense they are both synchronous, one by a clock and another by signals like start and stop while using async methods, the two bodies involved in the communication do not need to

Figure 1: Sample page 1

```
pip install requests pillow
```

This installs the two libraries *requests* and *pillow* for URL retrieval and image processing, respectively.

With that, our pre-requisites are done. We can now download the software and get going. There are two ways to do so. First, if you have *git* installed, you can clone the repository at <https://github.com/theSage21/handwritten> using the following command:

```
git clone https://github.com/theSage21/handwritten
```

The other method does not require *git*, though it does require some sort of unzipping tool like *Unzip* in Linux systems or *Win-zip* in Windows. To go this way, use the link <https://github.com/theSage21/handwritten/archives/master.zip> to download the Zip file. Once downloaded, you can unzip the file using the following command on Linux systems:

```
unzip handwritten.zip.
```

This will create a folder called *handwritten*, where the program will be.

Now, we navigate to the *handwritten* folder using the following command:

```
cd handwritten
```

We need to create a plain text file of what we need written. So let's use our favourite editor and open a blank *txt* file with the following command:

```
vim example.txt
or
gedit example.txt
or
notepad example.txt
```

Let's copy-paste our text into the file, and make sure that no line is longer than 100 characters.

Next, let's get the individual lines using the following command:

ready for data transfer: this is acceptable in situations where the polling interval is of such a size that it tends to minimize the number of times the device has to be checked. Attaining the value of this golden interval is a very difficult task and in practice it is usually under or over the perfect value. As a result the polling method ends up wasting a lot of time due to the constant checks.

Interrupt driven IO is when the IO device interrupts the software, leaving

Figure 2: Sample page 2

```
python get_hand.py example.txt 0
```

Here, the first part invokes Python and instructs it to run the script named *get_hand.py*. The variables given to the script are the name of the text file used to generate the text and the starting line number. If you do not provide the latter, it is assumed to be 0.

This will create a folder called *images* in the *handwritten* folder and put each line's handwritten text in there. The lines will be PNG images with the line number as names. You need not worry about this as it is an intermediate step in the process. In case some line does not seem good enough, you can always regenerate it by specifying the line number and stopping the script once it's done.

Now we need to run the other script in the folder called *make_page.py* using the following command:

```
python make_page.py 20
```

What 20 denotes is the number of lines to be used per page. In case it is omitted, the default value is 20. This script will create a folder called *pages* in the *handwritten* folder, and the pages of your text will be generated there as PNG images. They will be labelled with numbers denoting the page numbers.

If you would like to use this multiple times, you must make sure that the *images* and *pages* folders do not contain anything before running the script. If they do contain something, it will be included in the generation of the handwritten text.

As an example use case, I had submitted one of my assignments using this software. Some of the pages of that assignment are shown in Figures 1 and 2. 

References

- [1] Handwritten: <<http://theSage21.github.io/handwritten/>>
- [2] Issues may be reported at <<https://github.com/theSage21/handwritten/issues/>>

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